

# Passive photodiode visibility — run report

## RUN

190608\_pd-visibility\_pd-visibility

Status: completed | Started: 2026-09-23T18:06:08.812756+00:00

## PURPOSE

Measure fringe contrast on the selected PD input under external drive. No RP output initialization or PAX connection.

## REQUIRED OPTICAL SETUP

DC-coupled PD on the selected RP input; external drive must span complete fringes. Set its frequency and, when known, the blocked-light voltage offset. RP must already have the Pyrpl FPGA loaded.

## RECORDED BENCH CONTEXT (RECIPE)

PD: in2; external drive: 0.5 Hz; RP outputs unchanged

## OPERATOR SETUP NOTES

V<sub>pi</sub> expectation only; verify channel and measured driver gain before calibration.

## DATA

3 recorded samples; acquisition: 3

Source: data.csv; full acquisition settings: recipe.json; status: run.json

# Photodiode visibility

## PD INPUT

in2

## CAPTURES

3

## VALID CAPTURES

3

## VISIBILITY MEAN

0.02460719997707431

## VISIBILITY STD

0.0007221925699290588

## DARK VOLTAGE V

None

## EXTERNAL FREQUENCY HZ

0.5

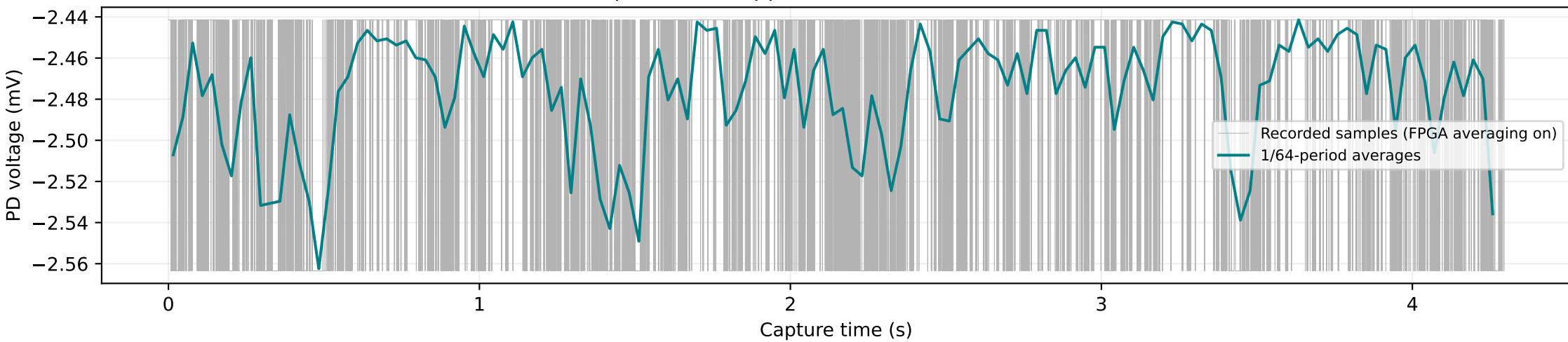
## DEFINITION

$(I_{\max} - I_{\min}) / (I_{\max} + I_{\min})$ ,  $I = \text{polarity} * (V - V_{\text{dark}})$ ; extrema of 1/64-period averages

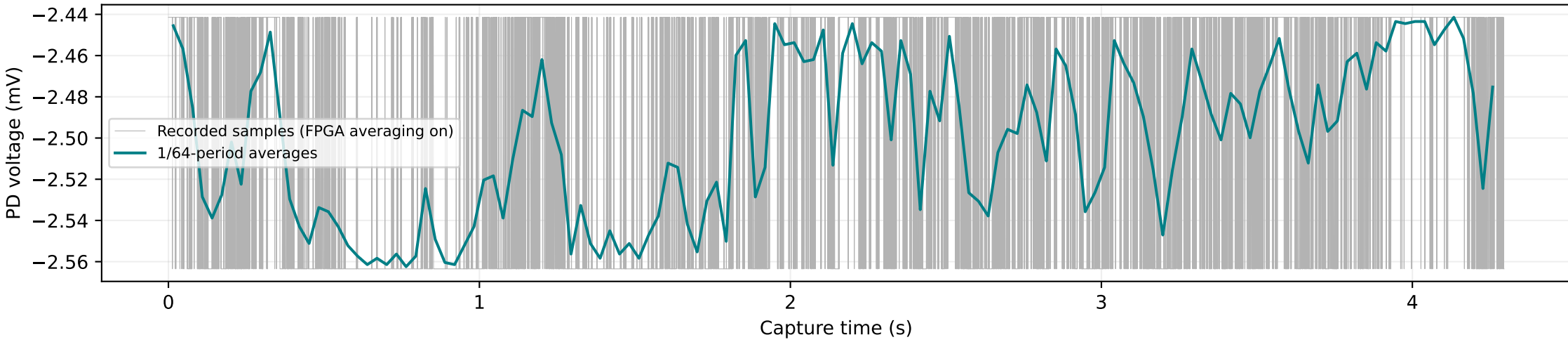
## INTERPRETATION

Observed contrast assumes DC coupling, linear unsaturated PD, and full fringe excursion. A missing dark measurement gives apparent contrast only. Scatter is not total uncertainty.

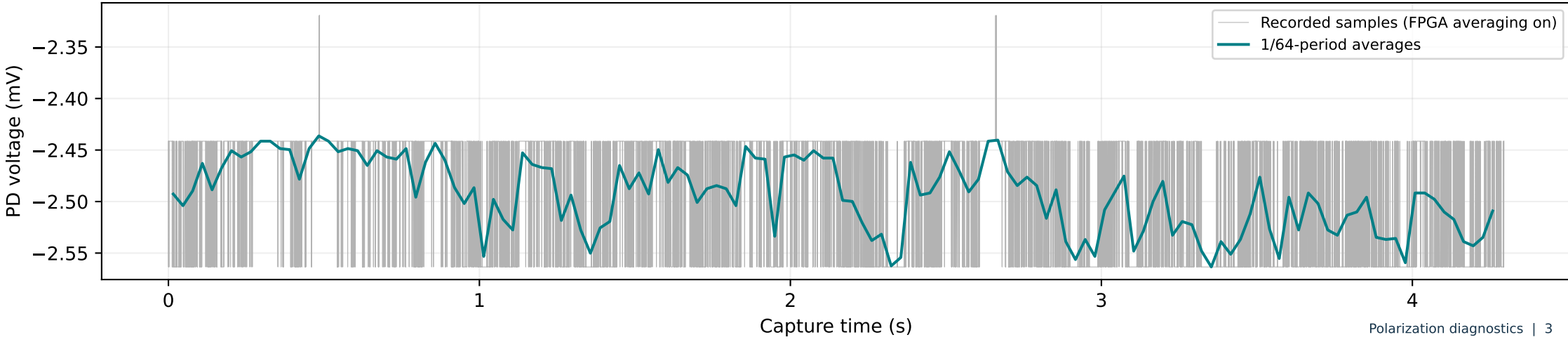
in2 / capture-000: apparent; zero dark offset assumed



in2 / capture-001: apparent; zero dark offset assumed



in2 / capture-002: apparent; zero dark offset assumed



# Active parameters

Passive acquisition; voltage at the selected PD input. No actuator commands.

| Parameter                 | Recorded value  |
|---------------------------|-----------------|
| label                     | "pd-visibility" |
| duration_s                | 12.0            |
| report                    | "pdf"           |
| pd_input                  | "in2"           |
| visibility_frequency_hz   | 0.5             |
| visibility_dark_voltage_v | null            |